CS404-Big Data Analytics
Project Abstract

Project Title: **Distributed Analytics of Global Torrent Activity**

Group Members :

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Objectives:
[+] To build a scalable big-data pipeline that ingests and processes BitTorrent ecosystem metadata (torrent catalogs, DHT discoveries, and tracker swarm statistics) across four media categories
[+] To quantify torrent activity volume, release-to-leak lag, and popularity decay curves for movies, TV shows, games, and music
[+] To analyze the impact of pricing, DRM, and streaming-platform exclusivity on torrent levels through joins with legitimate-market datasets (IMDb, Steam, Spotify charts)
[+] To visualize cross-media torrent sharing trends through interactive real-time dashboards

Abstract:
Torrent networks generate massive, publicly observable activity data, yet this activity is rarely measured systematically across media types. This project builds a big-data analytics platform that studies torrent activity across movies, TV shows, games, and music using only public BitTorrent metadata — no copyrighted content is downloaded. Input data includes the public RARBG catalog dump (~2.5 million torrents), live discoveries from a self-hosted DHT crawler, and hourly seeder/leecher statistics scraped from public trackers for 100,000+ active torrents, accumulating hundreds of millions of time-series records. Data flows through Kafka into HDFS as partitioned Parquet and is processed with Apache Spark, then joined with IMDb, Steam, and Spotify datasets using Hive. Expected outputs include a cross-media torrent activity index, release-to-availability lag analysis, popularity decay models, pricing and

platform-exclusivity impact studies, and interactive dashboards — demonstrating a complete batch-plus-streaming big data architecture.

Big Data Technologies to be used:
[+] Apache Kafka (streaming ingestion)
[+] HDFS + Apache Parquet (distributed storage)
[+] Apache Spark / PySpark + Spark Structured Streaming (processing)
[+] Apache Hive (warehousing and joins)
[+] Apache Airflow (orchestration)
[+] Apache Superset / Grafana (dashboards)
[+] Spark MLlib (trend and regression modelling)

Timelines (In weeks):
[+] W1-2: Environment setup; deploy DHT crawler and tracker scraper (data collection begins); load RARBG catalog
[+] W3-4: Kafka + HDFS ingestion pipeline; Spark ETL — deduplication, title parsing, category tagging
[+] W5-6: Integrate external datasets (IMDb, Steam, Spotify); build Hive warehouse
[+] W7-8: Analytics — torrent index, leak-lag, decay curves, pricing/exclusivity regression
[+] W9: Dashboards and visualization
[+] W10: Final report, testing, and demonstration
